



AKTU

B.Tech Sem-6

CS IT & CS Allied

CN: Computer Networks

ONE SHOT Revision

Unit-5 :Application Layer

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Computer Networks

Unit-5

AKTU Syllabus

Application Layer: ^{Imp} Domain Name System, World Wide Web and ^{HTTP} Hyper Text Transfer Protocol, ^{SMTP, POP3, IMAP Imp} Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography – basic concepts.

↓
TELNET

↓
SNMP

↓
RSA algo

AKTU PYQs

1. Explain DNS. (AKTU 2022-23, 2023-24, 2021-22, 2018-19, 2017-18, 2016-17)
2. Explain HTTP and its uses. (AKTU 2021-22, 2017-18)
3. State difference between HTTP and HTTPS (AKTU 2022-23)
4. "Elaborate about TELNET and its working procedure." (AKTU 2016-17, 2017-18, 2018-19, 2023-24)
5. What is the purpose of the Domain Name System? Discuss the three main divisions of the domain name space. (AKTU 2018-19)
6. Explain SNMP in detail. (AKTU 2023-24)
7. Differentiate between FTP and HTTP. (AKTU 2023-24)
8. Explain the following terms:
9. (i) FTP (ii) SMTP (AKTU 2022-23, 2021-22)
10. Define SNMP Protocols and working scenario (AKTU 2022-23)

...AKTU PYQs

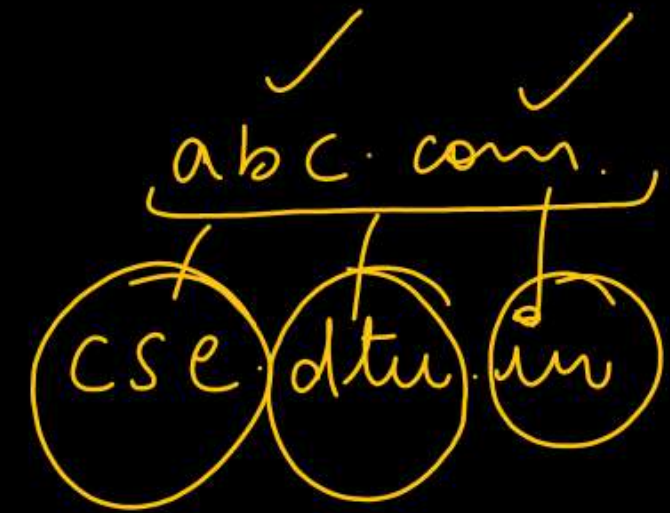
11. Define Compression. (AKTU 2023-24)
12. Explain Cryptography. Write about RSA Algorithm with example. (AKTU 2023-24)
13. Explain Asymmetric cryptography. Also, write the steps used in RSA algorithm, demonstrate the transmission of character "F" using RSA. (AKTU 2022-23)
14. List out few email gateways. (AKTU 2016-17)
15. (i) How is TFTP different from FTP?
(ii) What three functions can SNMP perform to manage network devices? (AKTU 2018-19)
16. What is the difference between a user agent (UA) and a mail transfer agent (MTA)? (AKTU 2018-19)

Domain Name System (DNS)

- DNS translates human-readable domain names (e.g., www.google.com) into machine-readable IP addresses (e.g., 142.250.190.14), enabling internet communication.
- It enables computers to locate and communicate with each other on the internet.
- Functions as a hierarchical, distributed database.

- Queries pass through multiple levels:

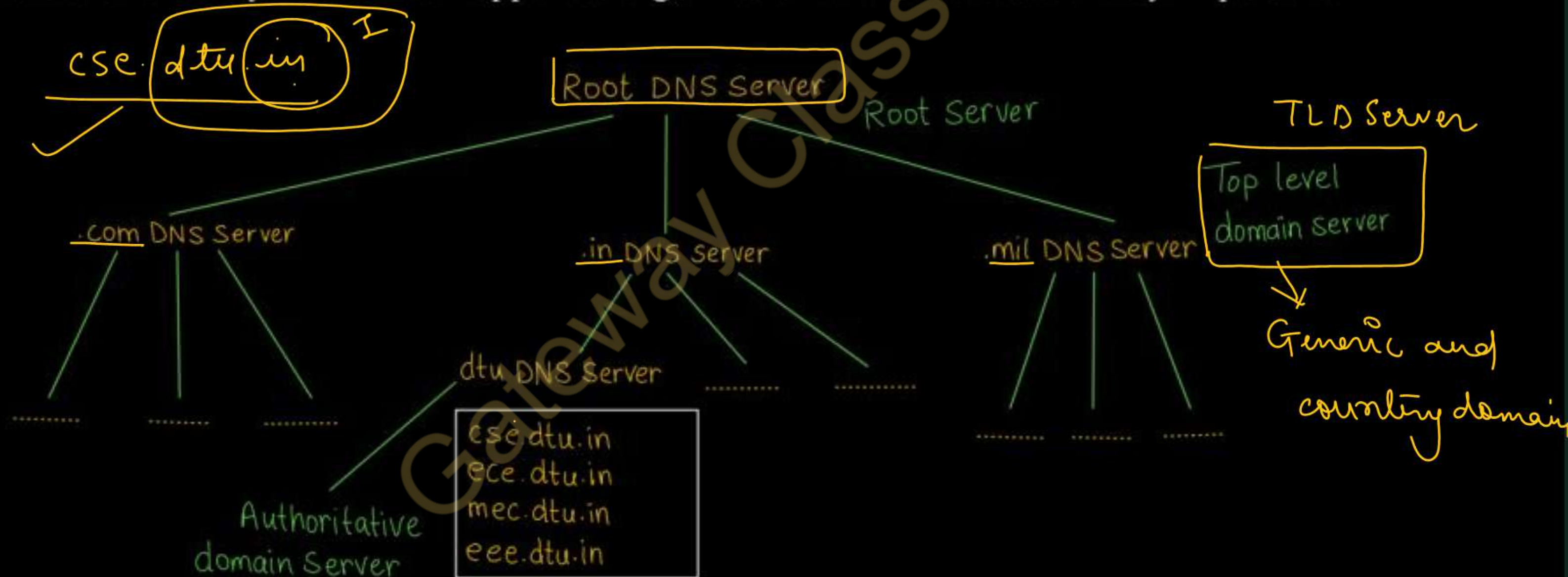
- Root server (13)
- Top-Level Domain (TLD) server
- Authoritative server (stores the specific IP address).



- Ensures seamless website access using easy-to-remember names instead of numerical IP addresses.

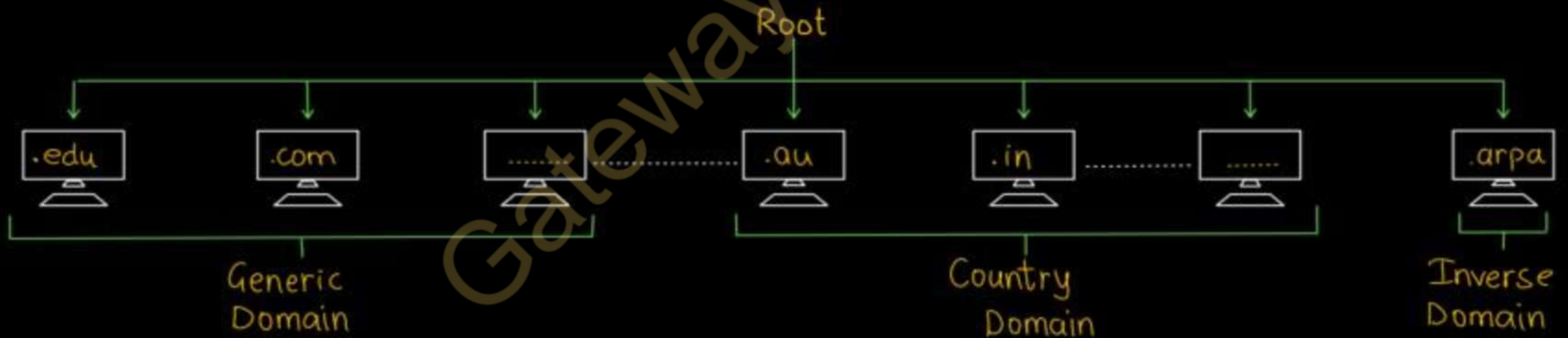
Structure of DNS

- It is very difficult to find out the IP address associated with a website as there are millions of websites and we should be able to generate the IP address for all of them immediately, there should not be a lot of delays for that to happen so organization of the database is very important.



Types of Domain

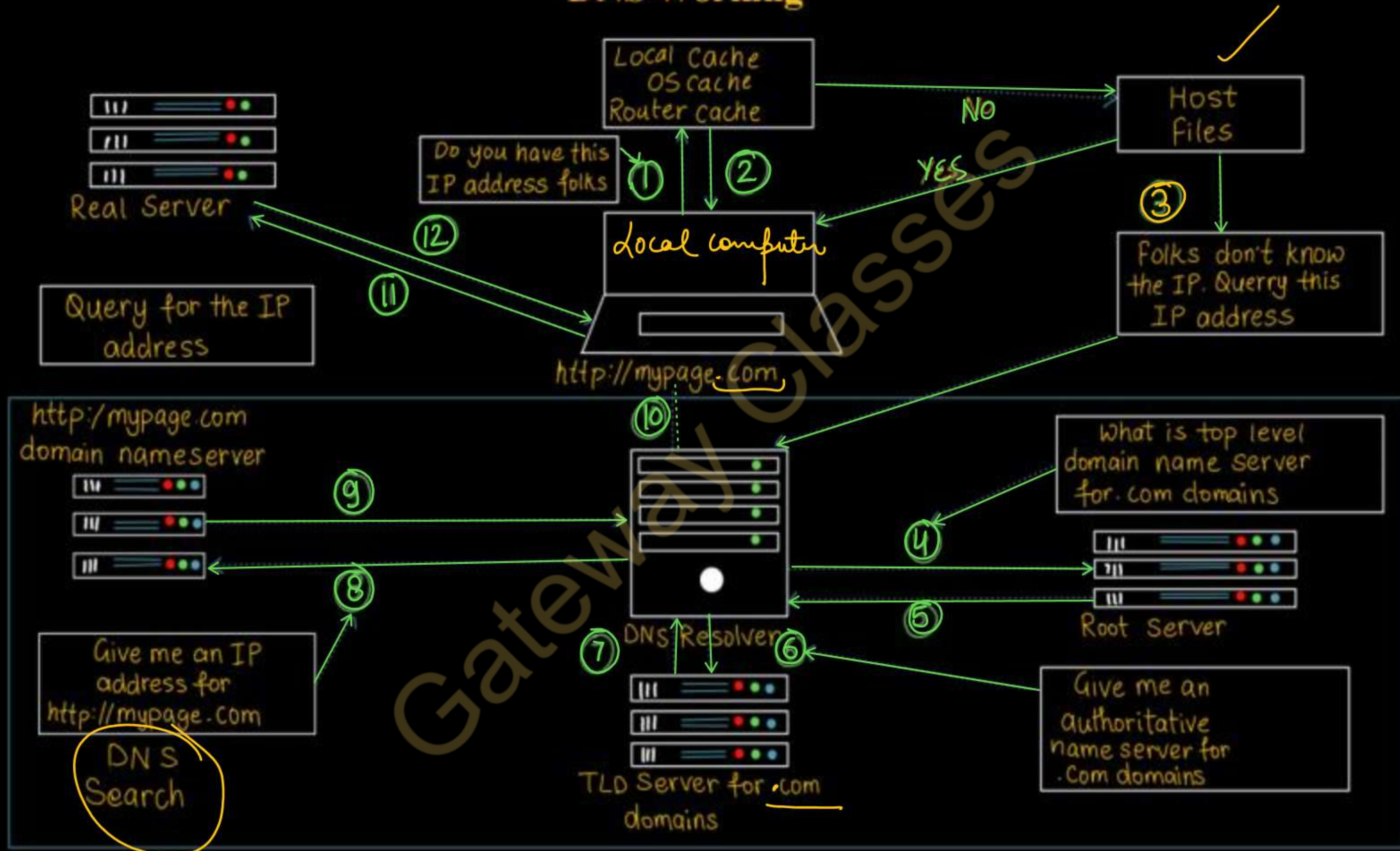
- **Generic Domains:** .com(commercial), .edu(educational), .mil(military), .org(nonprofit organization), .net(similar to commercial). All these are generic domains.
- **Country Domain:** .in (India) .us .uk
- **Inverse Domain:** if we want to know what is the domain name of the website. IP to domain name mapping. So, DNS can provide both the mapping.
- To find IP addresses of **www.hello.org**, we have to type **nslookup www.hello.org**



DNS Working

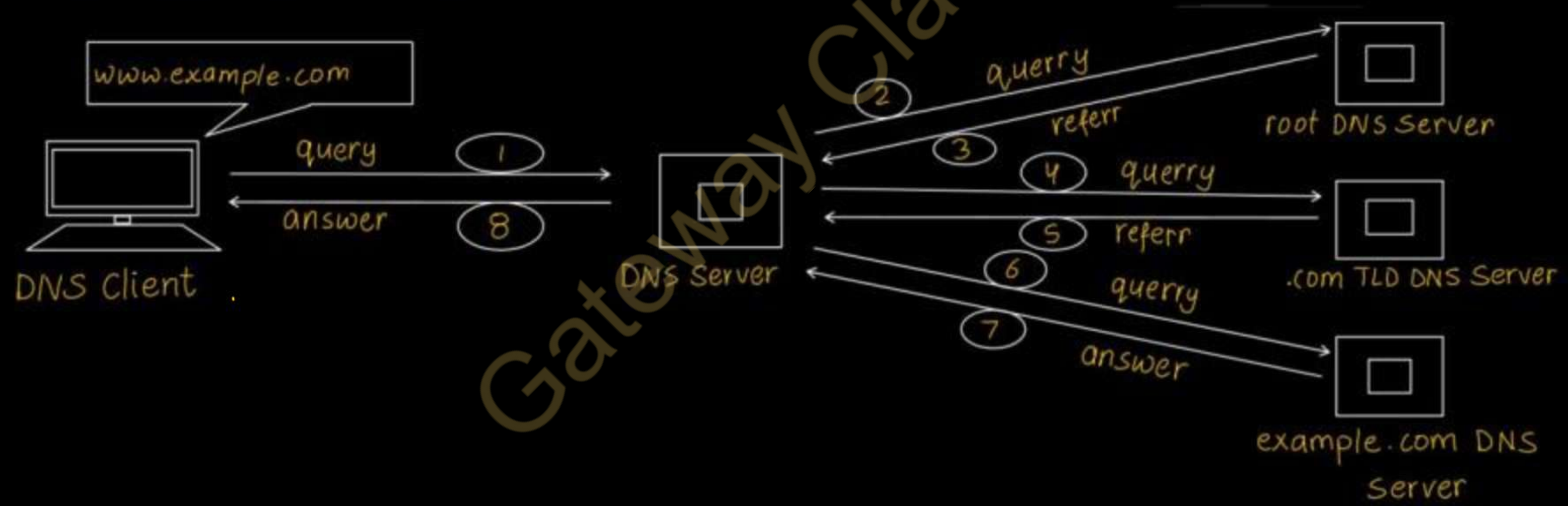
- We type a website like <https://mypage.com> in our browser, our computer tries to find the IP address.
- First, it checks local cache (our browser, OS, or router) to see if it already knows the IP address.
- If local cache doesn't have IP, check host files (used for specific manual mappings).
- If still not found, the query is sent to a **DNS resolver** (usually provided by your ISP).
- DNS resolver sends query to a **Root DNS server**, which points to Top-level domain (TLD) server (.com server for this example).
- **TLD server** then directs the resolver to the authoritative name server for **mypage.com**.
- **Authoritative nameserver** knows the exact IP address for **mypage.com** and sends it back to the resolver. This server holds the actual DNS records and responds with the IP address.
- Resolver passes the IP address to our computer.
- Our computer uses the IP address to connect to the real server where the website is hosted.
- The website loads in our browser.

DNS Working



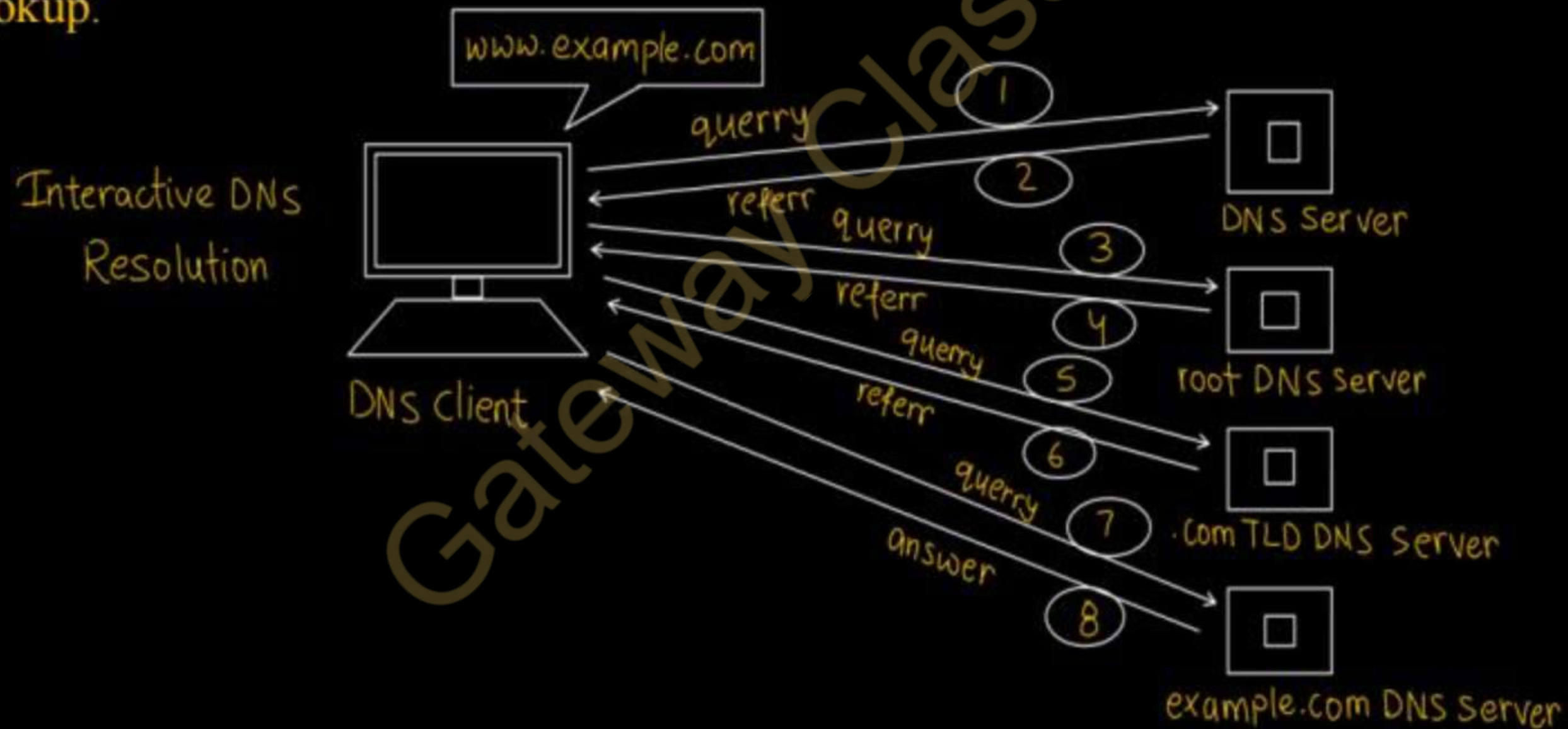
DNS recursive query:

- DNS server, who received your query will do all job of fetching the answer, and giving it back to you.
- During this process, the DNS server might also query other DNS servers in the internet on your behalf, for the answer. **One DNS server communicates with several other DNS servers to find an IP address and return it to the client.**



Iterative DNS query:

- In this, the DNS server, will not go and fetch the complete answer for your query, but will give back a referral to other DNS server's, which might have the answer.
- In an iterative **DNS query**, where the client communicates directly with each **DNS** server involved in the lookup.



Caching:

- When a server asks for a mapping from another server and receives the response, it stores this information in its cache memory before sending it to the client.
- If the same or another client asks for the same mapping, it can check its cache memory(server marks the response as *unauthoritative*)and solve the problem.

Problems:

If a server caches a mapping for a long time, it may send an outdated mapping to the client.

To counter this, the authoritative server adds TTL to the mapping.

Importance of DNS

➤ **User-Friendly Navigation:**

- DNS simplifies user navigation by allowing users to access websites using easy-to-remember names instead of dealing with complex numerical addresses.

➤ **Load Balancing:**

- Help distribute traffic across multiple servers and ensures efficient load management and enhances website performance during traffic periods.

➤ **Email Routing:**

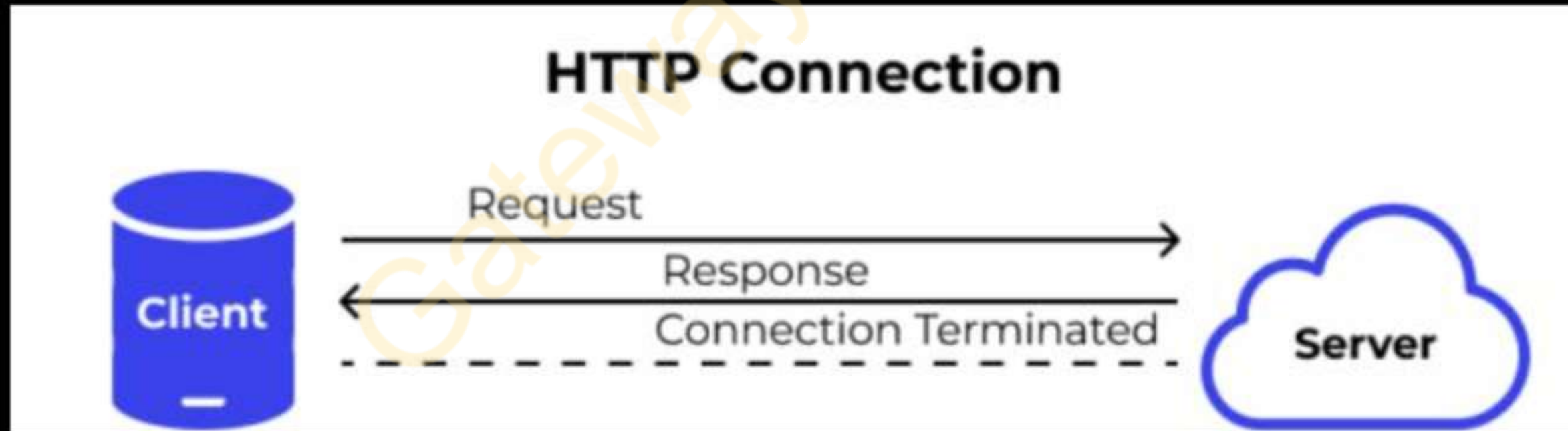
- Enable email routing by directing emails to the correct mail servers associated with a domain.

➤ **Security Features:** adds extra layer of security ensuring DNS responses are authentic not tampered.

➤ **Scalability:** DNS provides scalability by allowing organizations to add subdomains or new services without significant changes to their overall infrastructure.

HTTP (Hypertext Transfer Protocol)

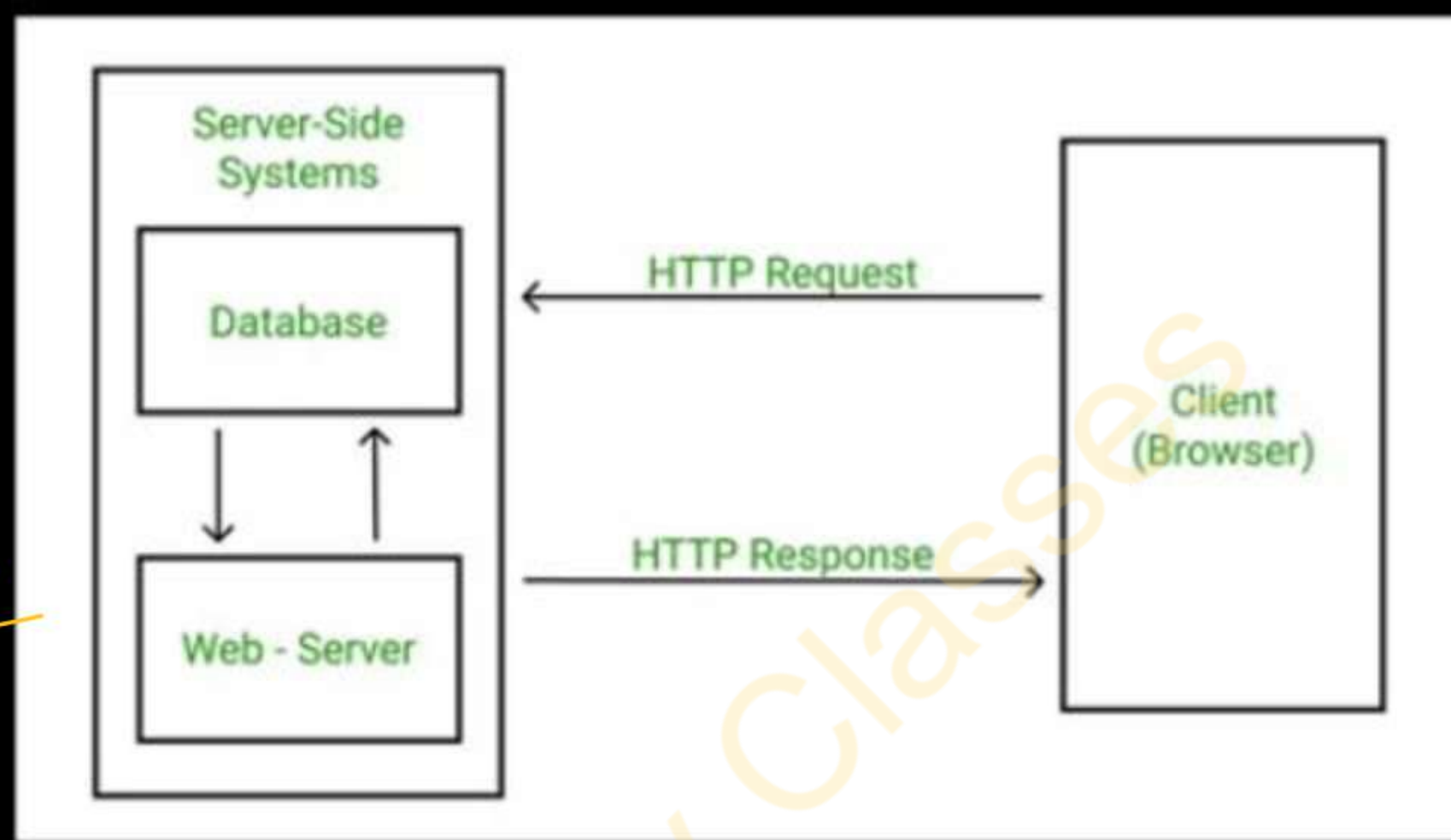
- It is a fundamental protocol of the Internet, enabling data transfer between a client and a server.
- It is the foundation of data communication for the World Wide Web. port no. 80
- HTTP provides a standard between a web browser and a web server to establish communication.
- A set of rules for transferring data (text, images, and multimedia files) from one computer to another.
- Whenever a web user opens their web browser, the user indirectly uses HTTP.



Methods of HTTP

- **GET**: Used to retrieve data from a specified resource.
- **POST**: Used to submit data to be processed by a specified resource.
- **PUT**: Used to update or create a resource on the server.
- **PATCH**: Similar to PUT but used for partial modifications to a resource.
- **DELETE**: Used to remove a specified resource from the server.
- **HEAD**: Similar to GET but retrieves only the response headers, useful for checking resource properties without transferring the full content.
- **OPTIONS**: Used to retrieve the communication options available for a resource, including supported methods and headers.
- **TRACE**: Used for debugging purposes to echo the received request back to the client.
- **CONNECT**: Used to establish a tunnel to the server through an HTTP proxy, commonly used for SSL/TLS connections.

80



Features:

- Stateless: Each request is independent, and the server doesn't retain previous interaction information.
- **Text-Based**: Messages are in plain text, making them readable and debuggable.
- **Client-Server Model**: Follows a client-server architecture for requesting and serving resources.
- **Request-Response**: Operates on a request-response cycle between clients and servers.
- **Request Methods**: Supports methods like GET, POST, PUT, DELETE for different actions.

Difference Between HTTP and HTTPS (AKTU)

HTTP(HyperText Transfer Protocol)	HTTPS(HyperText Transfer Protocol Secure)
URL begins with "http://".	In HTTPs, URL starts with "https://".
uses port number 80 for communication.	uses 443 port number for communication.
unsecure.	HTTPs is considered as secure.
works at Application Layer.	HTTPS works at Transport Layer.
Encryption is absent.	Encryption is present in HTTPS.
faster than HTTPS	slower than HTTP
Does not use data hashtags to secure data.	While HTTPS will have the data before sending it and return it to its original state on the receiver side.
In HTTP Data is transfer in plaintext.	In HTTPS Data transfer in ciphertext.
HTTP Does not require SSL/TLS or Certificates	Requires SSL/TLS implementation with Certificates.

Hello → plaintext
 ↓
 IFMMP → ciphertext

Advantages:

- Platform independence: Works on any operating system
- Compatibility: Compatible with various protocols and technologies
- Efficiency: Optimized for performance
- Security: Supports encryption for secure data transfer

Disadvantages:

- Lack of security: Vulnerable to attacks like man in the middle
- Performance issues: Can be slow for large data transfers
- Statelessness: Requires additional mechanisms for maintaining state

Remote Logging

- Remote Login is a process in which user can login into remote computer and use services that are available on the remote computer. It is implemented using Telnet.

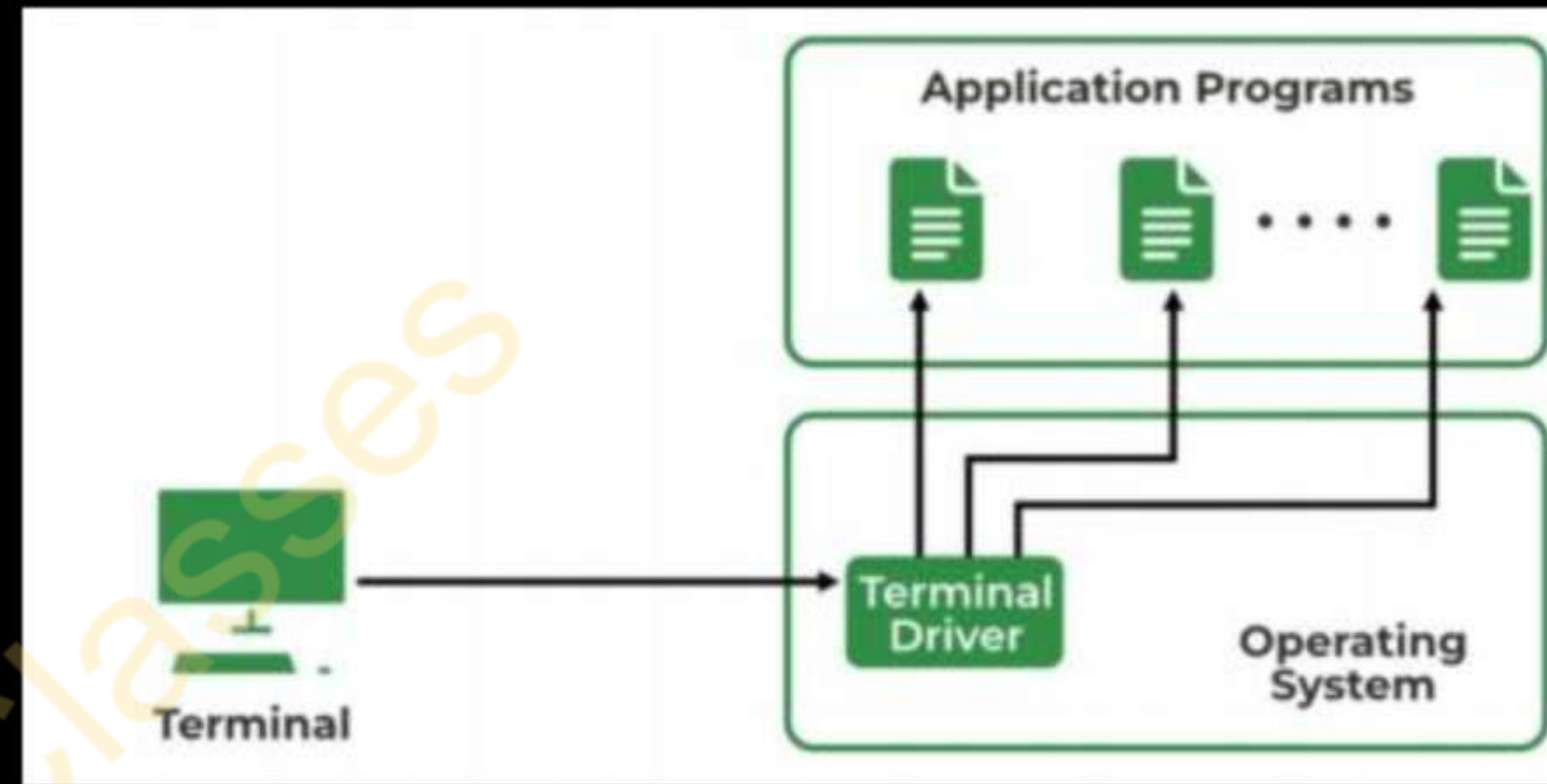
Example:

Students may want to connect to their university computer lab from their home to access application programs for doing homework assignments.

- One way is to create a client/server application program for each desired service using FTP or SMTP but it would be impossible to write a specific client/server program for each demand.
- Better solution is remote login that lets a user access any application program on a remote computer. After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.

Local Login:

- When a user logs into a local timesharing system, it types at terminal, keystrokes are accepted by terminal driver.
- Terminal driver passes the characters to the OS.
- OS interprets the combination of characters and invokes the desired application program or utility.



TELNET (TERminaL NETwork):

- It is the standard TCP/IP protocol for virtual terminal service proposed by ISO.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- TELNET is a general-purpose client/server application program.